



SIMCOM_A7600_Linux_USB_User_Guide _V1.01.00

LTE Module

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Version History

Version	Date	Chapter	What is new
V1.00.00	2019-06-06		New version
V1.01.00	2019-08-07	3.1、3.2	1. Add +DIALMODE and \$MYCONFIG description. 2. Replace /DEV/ttyUSB3 with /DEV/ttyUSB2.
V1.02.00	2020-11-27	3.3	3. Add +USBNETIP
	2021-08-10	2	4. Add NMEA port

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1 Introduction

1.1. Scope

This user guide serves the following purpose:

- Short introductions how to customize the USB driver for Simcom A7600 module in Linux OS
- Describes how software developers can use Linux devices for typical use cases.

1.2. Related Documents

1: A7600 ATC.

2 Device Driver Installation

In order to recognize the modem, you must add VID and PID in Linux driver option, after the operating system recognizes the modem, devices named /dev/ttyUSBx are created, for example:

- /dev/ttyUSB0 diag port for output developing messages
- /dev/ttyUSB1 AT port for AT commands
- /dev/ttyUSB2 Modem port for ppp-dial
- /dev/ttyUSB3 NMEA port for GNSS(GNSS must be supported)

2.1. Add VID and PID

1.Edit option.c source file

- a. gedit linux-x.x.x/drivers/usb/serial/option.c
- b. Check if the following #define statement exists,If the #define does not exist ,add:

```
#define SIMCOM_VENDOR_ID      0x1e0e
#define SIMCOM_PRODUCT_PID_X9011      0x9011
```

- c. Check the following struct statement exists,If the struct does not exist ,add the following struct

```
static const struct option_blacklist_info simcom_pid9011_blacklist = {
    .reserved = BIT(0) | BIT(1),
}
```

- d. Add the following line to usb_device_id option_ids[] structure, if not exists

```
{ USB_DEVICE(SIMCOM_VENDOR_ID, SIMCOM_PRODUCT_PID_X9011),
    .driver_info = (kernel_ulong_t)&simcom_pid9011_blacklist }
```

If your kernel code is different from the above, please be careful to modify it as appropriate.

2.2. Building a Linux Driver Module

How to compile and install a kernel module in Linux , follow the steps below will guide you along in compiling and install your option driver On Ubuntu operating system.

Step 1: Enter to kernel directory.

```
cd <your kernel directory>
```

Step 2: Build the driver.

```
sudo make -C /lib/modules/`uname -r`/build M=`pwd`/drivers/usb/serial
obj-m=option.o modules
```

Step 3: Load the driver and reboot.

```
sudo cp drivers/usb/serial/option.ko /lib/modules/`uname
-r`/kernel/drivers/usb/serial
sudo depmod
sudo reboot
```

2.3. Kernel Compilation Configuration

2.3.1. Compilation Configuration for USB Serial Driver

Configuration	Configuration(Y/N)
CONFIG_USB_SERIAL	Y
CONFIG_USB_SERIAL_OPTION	Y

2.3.2. Compilation Configuration for RNDIS Driver

Configuration	Configuration(Y/N)
CONFIG_USB_SERIAL	Y
CONFIG_USB_SERIAL_OPTION	Y
CONFIG_USB_USBNET	Y
CONFIG_USB_NET_CDCETHER	Y

2.3.3. Compilation Configuration for PPP Driver

Configuration	Configuration(Y/N)
CONFIG_USB_SERIAL	Y
CONFIG_USB_SERIAL_OPTION	Y
CONFIG_PPP	Y
CONFIG_PPP_FILTER	Y
CONFIG_PPP_MULTILINK	Y
CONFIG_PPP_BSDCOMP	Y
CONFIG_PPP_ASYNC	Y
CONFIG_PPP_SYNC_TTY	Y
CONFIG_PPP_DEFLATE	Y

3 Modem Usage

This chapter mainly introduces several commonly used dialing methods and their general processes.

USB devices must be recognized before use modem.

3.1. Description of AT Command

3.1.1 AT+DAILMODE Enable/Disable Auto-Dial

AT+DAILMODE Enable/Disable Auto-Dial	
Test Command AT+DAILMODE=?	Response +DAILMODE: (0-1) OK
Read Command AT+ DAILMODE?	Response + DAILMODE: 0 + DAILMODE: 1 OK
Write Command AT+ DAILMODE =<0/1>	Response a)If successfully: OK b)If failed: ERROR
Defined Values	
<0/1>	The Auto Dial status Enable/Disable, the default value is 1. 0 –Enable Auto Dail 1 –Disable Auto Dail The function will take effect immediately.

3.1.2 AT\$MYCONFIG Set RNDIS/ECM Mode

AT\$MYCONFIG Set RNDIS/ECM Mode	
Write Command AT\$MYCONFIG= " USBNETMODE " , <0/1>	Response a)If successfully: OK b)If failed: ERROR
Defined Values <0/1>	The RNDIS/ECM mode, the default value is 0. <u>0</u> –RNDIS <u>1</u> –ECM The function will reset modem then take effect.

NOTE

this command will auto reboot(take effect after reboot).

3.1.3 AT+USBNETIP Change RNDIS/ECM Private IP to Public IP

AT+USBNETIP Change RNDIS/ECM Private IP to Public IP	
Test Command AT+USBNETIP=?	Response +USBNETIP: (0-1)[,(0-255)][,(0-255)][,(1-254)][] OK
Write Command AT+USBNETIP=mode[,tpos[,dhcps[,thcp e]]]	Response a)If successfully: OK b)If failed: +CME ERROR: unknown error
Read Command AT+USBNETIP?	Response a)If successfully: +USBNETIP: mode OK b)If failed: +CME ERROR: unknown error
Parameter Saving Mode	AUTO_SAVE

Defined Values

<mode>	0 –Public Ip 1 –Private Ip
<tpos>	The third position of Gateway Addr
<dhcps>	Dhcp start value
<dhcpe>	Dhcp end value

Examples:

Windows 打开 cmd 输入 ipconfig

以太网适配器 以太网 7:

```

连接特定的 DNS 后缀 . . . . . :
IPv6 地址 . . . . . : 2409:8960:1e58:324c:2934:bda5:f9e8:88d7
临时 IPv6 地址. . . . . : 2409:8960:1e58:324c:65cd:269c:30d1:17f0
本地链接 IPv6 地址. . . . . : fe80::2934:bda5:f9e8:88d7%18
IPv4 地址 . . . . . : 192.168.0.100
子网掩码 . . . . . : 255.255.255.0
默认网关. . . . . : fe80::1234%18
                    192.168.0.1

```

串口输入 AT+USBNETIP=0,10,117, 成功后, 再次输入 **ipconfig**

以太网适配器 以太网 7:

```

连接特定的 DNS 后缀 . . . . . :
IPv6 地址 . . . . . : 2409:8960:1e58:324c:2934:bda5:f9e8:88d7
临时 IPv6 地址. . . . . : 2409:8960:1e58:324c:65cd:269c:30d1:17f0
本地链接 IPv6 地址. . . . . : fe80::2934:bda5:f9e8:88d7%18
IPv4 地址 . . . . . : 192.168.10.117
子网掩码 . . . . . : 255.255.255.0
默认网关. . . . . : fe80::1234%18
                    192.168.10.1

```

串口输入 AT+USBNETIP=0,12,98, 成功后, 再次输入 **ipconfig**

以太网适配器 以太网 7:

```

连接特定的 DNS 后缀 . . . . . :
IPv6 地址 . . . . . : 2409:8960:1e58:324c:2934:bda5:f9e8:88d7
临时 IPv6 地址. . . . . : 2409:8960:1e58:324c:65cd:269c:30d1:17f0
本地链接 IPv6 地址. . . . . : fe80::2934:bda5:f9e8:88d7%18
IPv4 地址 . . . . . : 192.168.12.98
子网掩码 . . . . . : 255.255.255.0
默认网关. . . . . : fe80::1234%18
                    192.168.12.1

```

3.2. Test AT Commands

```
#cat /dev/ttyUSB2 &
#echo -e "at\r\n">/dev/ttyUSB2
#
OK
```

3.3. Use PPP Data connection

3.3.1. How Does a PPP Dial-Up Connection Work?

You will need the right software and a couple of pieces of information before you start. First, check the pppd. If the programs do not exist, you can download the source code from <https://ppp.samba.org/download.html> and port them to your embedded development environment. Next you must write configuration file for pppd.

3.3.1.1. Chat Scription

```
#named simcom-connect-chat and place in /etc/ppp/peers
ABORT "BUSY"
ABORT "NO CARRIER"
ABORT "NO DIALTONE"
ABORT "ERROR"
ABORT "NO ANSWER"
TIMEOUT 30
"" AT
OK ATE0
OK ATI;+CSUB;+CSQ;+CPIN?;+COPS?;+CGREG?;&D2
# Insert the APN provided by your network operator, default apn is 3gnet
OK AT+CGDCONT=1,"IP","3gnet",,0,0
OK ATD*99#
CONNECT
```

```
#named simcom-disconnect-chat and place in /etc/ppp/peers
ABORT "ERROR"
```

```
ABORT "NO DIALTONE"
SAY "\nSending break to the modem\n"
"" +++
"" +++
"" +++
SAY "\nGoodbay\n"
```

3.3.1.2. Configure dialing and AT port

```
# named simcom-pppd and place in /etc/ppp/peers
/dev/ttyUSB2 115200

#Insert the username and password for authentication, default user and password are
test

user "test" password "test"

# The chat script, customize your APN in this file
connect 'chat -s -v -f /etc/ppp/peers/simcom-connect-chat'

# The close script
disconnect 'chat -s -v -f /etc/ppp/peers/simcom-disconnect-chat'

# Hide password in debug messages
hide-password

# The phone is not required to authenticate
noauth

# Debug info from pppd
debug

# If you want to use the HSDPA link as your gateway
defaultroute

# pppd must not propose any IP address to the peer
noipdefault

# No ppp compression
novj
novjccomp
noccip
ipcp-accept-local
```

```
ipcp-accept-remote
local
# For sanity, keep a lock on the serial line
lock
modem
dump
nodetach
# Hardware flow control
nocrtscts
remotename 3gppp
ipparam 3gppp
ipcp-max-failure 30
# Ask the peer for up to 2 DNS server addresses
usepeerdns
```

3.3.1.3. Dial-Up Connection

```
# pppd call simcom-pppd &
```

When you see the output below, it shows that dial-up succeeded.

```
Connect: ppp0 <--> /dev/ttyUSB2
sent [LCP ConfReq id=0x1 <asynctest 0x0> <magic 0x5107d141> <pcomp> <accomp>]
rcvd [LCP ConfReq id=0x0 <asynctest 0x0> <auth chap MD5> <magic 0x9a5c1936> <pcomp>
<accomp>]
sent [LCP ConfAck id=0x0 <asynctest 0x0> <auth chap MD5> <magic 0x9a5c1936> <pcomp>
<accomp>]
rcvd [LCP ConfAck id=0x1 <asynctest 0x0> <magic 0x5107d141> <pcomp> <accomp>]
sent [LCP EchoReq id=0x0 magic=0x5107d141]
rcvd [LCP DiscReq id=0x1 magic=0x9a5c1936]
rcvd [CHAP Challenge id=0x1 <dd93b9f04d75e2bbba3786f6d24df3d7>, name =
"UMTS_CHAP_SRVR"]
sent [CHAP Response id=0x1 <498d4d7cf3b59dacfc07a45ce6eb7e26>, name = "test"]
rcvd [LCP EchoRep id=0x0 magic=0x9a5c1936 51 07 d1 41]
rcvd [CHAP Success id=0x1 ""]
```

```
CHAP authentication succeeded
CHAP authentication succeeded
sent [IPCP ConfReq id=0x1 <addr 0.0.0.0> <ms-dns1 0.0.0.0> <ms-dns2 0.0.0.0>]
rcvd [IPCP ConfReq id=0x0]
sent [IPCP ConfNak id=0x0 <addr 0.0.0.0>]
rcvd [IPCP ConfNak id=0x1 <addr 10.51.68.23> <ms-dns1 222.66.251.8> <ms-dns2
116.236.159.8>]
sent [IPCP ConfReq id=0x2 <addr 10.51.68.23> <ms-dns1 222.66.251.8> <ms-dns2
116.236.159.8>]
rcvd [IPCP ConfReq id=0x1]
sent [IPCP ConfAck id=0x1]
rcvd [IPCP ConfAck id=0x2 <addr 10.51.68.23> <ms-dns1 222.66.251.8> <ms-dns2
116.236.159.8>]
Could not determine remote IP address: defaulting to 10.64.64.64
local IP address 10.51.68.23
remote IP address 10.64.64.64
primary DNS address 222.66.251.8
secondary DNS address 116.236.159.8
Script /etc/ppp/ip-up started (pid 6616)
Script /etc/ppp/ip-up finished (pid 6616), status = 0x0
```

Now PPP call is set up successfully. Please use following commands to check IP/DNS/Route.

```
# ifconfig ppp0
ppp0      Link encap:Point-to-Point Protocol
          inet addr:10.216.159.39 P-t-P:10.64.64.64 Mask:255.255.255.255
          UP POINTOPOINT RUNNING NOARP MULTICAST MTU:1500 Metric:1
          RX packets:9 errors:0 dropped:0 overruns:0 frame:0
          TX packets:9 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:3
          RX bytes:362 (362.0 B) TX bytes:316 (316.0 B)

# cat /etc/resolv.conf
nameserver 221.180.132.108

# route -n
```

Kernel IP routing table

Destination	Gateway	Genmask	Flags	Metric	Ref	Use	Iface
0.0.0.0	0.0.0.0	0.0.0.0	U	0	0	0	ppp0
10.64.64.64	0.0.0.0	255.255.255.255	UH	0	0	0	ppp0

```
# ping baidu.com
```

```
PING baidu.com (220.181.57.216) 56(84) bytes of data.
```

```
64 bytes from 220.181.57.216: icmp_seq=1 ttl=50 time=84.0 ms
```

```
64 bytes from 220.181.57.216: icmp_seq=2 ttl=50 time=34.2 ms
```

Following commands can be used to terminate PPPD process to disconnect a PPP call:

```
# killall pppd
```

3.4. Use ECM Data connection

Enable ECM

```
# cat /dev/ttyUSB2 &
```

```
# echo -e "AT+DIALMODE=0\r\n">/dev/ttyUSB2
```

```
#
```

```
# OK
```

```
# echo -e "AT+$MYCONFIG=\"usbnetmode\",1 ">/dev/ttyUSB2
```

```
#
```

```
# OK
```

Please use following commands to check IP/DNS/Route.

```
# ifconfig usb0
```

```
usb0      Link encap:Ethernet  HWaddr ae:68:46:d6:b2:80
          inet addr:192.168.0.100  Bcast:192.168.0.255  Mask:255.255.255.0
          inet6 addr: fe80::ac68:46ff:fed6:b280/64  Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:45 errors:0 dropped:0 overruns:0 frame:0
          TX packets:104 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4237 (4.2 KB)  TX bytes:13148 (13.1 KB)
```

```
# ping baidu.com
PING baidu.com (123.125.114.144) 56(84) bytes of data.
64 bytes from 123.125.114.144: icmp_seq=1 ttl=56 time=114 ms
64 bytes from 123.125.114.144: icmp_seq=2 ttl=56 time=58.6 ms
64 bytes from 123.125.114.144: icmp_seq=3 ttl=56 time=45.1 ms
```

3.5. Use RNDIS Data connection

```
# cat /dev/ttyUSB2 &
# echo -e "AT+DIALMODE=0\r\n"/dev/ttyUSB2
#
# OK
# echo -e "AT+MYCONFIG=\"usbnetmode\",0\r\n"/dev/ttyUSB2
#
# OK
```

Please use following commands to check IP/DNS/Route.

```
# ifconfig usb0
usb0      Link encap:Ethernet  HWaddr ae:68:46:d6:b2:80
          inet addr:192.168.0.100  Bcast:192.168.0.255  Mask:255.255.255.0
          inet6 addr: fe80::ac68:46ff:fed6:b280/64  Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:45 errors:0dropped:0 overruns:0 frame:0
          TX packets:104 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4237 (4.2 KB)  TX bytes:13148 (13.1 KB)

# ping baidu.com
PING baidu.com (220.181.38.148) 56(84) bytes of data.
64 bytes from 220.181.38.148: icmp_seq=1 ttl=50 time=94.8 ms
64 bytes from 220.181.38.148: icmp_seq=2 ttl=50 time=135 ms
64 bytes from 220.181.38.148: icmp_seq=3 ttl=50 time=61.9 ms
```

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4 Troubleshooting

If Linux does not create devices, check for the kernel module:

```
# lsmod | grep option
```

If entries aren't found, load the kernel module with root privileges:

```
# modprobe option
```

Check dmesg output to see that the radio was detected:

```
# dmesg | grep option
```

Check dmesg output to see that the radio was detected:

```
# dmesg | grep option

[ 16.672003] usbcore: registered new interface driver option
[ 16.672105] option 2-1.2:1.0: GSM modem (1-port) converter detected
[ 16.672216] option 2-1.2:1.1: GSM modem (1-port) converter detected
[ 16.672292] option 2-1.2:1.2: GSM modem (1-port) converter detected
[ 16.672365] option 2-1.2:1.3: GSM modem (1-port) converter detected
[ 16.672438] option 2-1.2:1.4: GSM modem (1-port) converter detected
```

If this returns an error response, the kernel module is not on your system. You will need to build the driver

5 Appendix A Abbreviations

Table 1: Terms and Abbreviations

Abbreviation	Description
USB	Universal Serial Bus
PPP	Point-to-Point Protocol. The Point-to-Point Protocol is designed for simple links which transport packets between two ports. These links provide full-duplex simultaneous bi-directional operation, and are assumed to deliver packets in order. It is intended that PPP provides a common solution for easy connection of a wide variety of hosts, bridges and routers.
IPCP	IP Control Protocol
IP	Internet Protocol
DNS	Domain Name Server